Data format “2023_dx_Leaf_Spec_Tomato_Stress”

The data set is organized in two different structures:

<table>
<tr>
<td>PAR = 300

In this example:

35 = 3 treatments x 5 replicates x between 2-3 samples

832 = number of spectral samples</td>
<td>
<table>
<tr><th>Field</th><th>Value</th></tr>
<tr><td>wvl</td><td>35x832 double</td></tr>
<tr><td>PAR</td><td>35x832 double</td></tr>
<tr><td>PARf</td><td>35x832 double</td></tr>
<tr><td>refl_real</td><td>35x832 double</td></tr>
<tr><td>trans_real</td><td>35x832 double</td></tr>
<tr><td>abs_real</td><td>35x832 double</td></tr>
<tr><td>Fup</td><td>35x832 double</td></tr>
<tr><td>Fdw</td><td>35x832 double</td></tr>
<tr><td>treatment</td><td>35x1 string</td></tr>
<tr><td>repli</td><td>35x1 string</td></tr>
<tr><td>sample</td><td>35x1 string</td></tr>
<tr><td>refl_real_smooth</td><td>35x832 double</td></tr>
<tr><td>trans_real_smooth</td><td>35x832 double</td></tr>
<tr><td>abs_real_smooth</td><td>35x832 double</td></tr>
</table>
</td>
</tr>
<tr>
<td>PAR = 1000

In this example:

35 = 3 treatments x 5 replicates x between 2-3 samples

832 = number of spectral samples</td>
<td>
<table>
<tr><th>Field</th><th>Value</th></tr>
<tr><td>wvl</td><td>35x832 double</td></tr>
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</table>
</td>
</tr>
</table>

Where for PAR = 300 and PAR = 1000:

wvl – wavelength (nm)

PAR – total incoming radiance spectrum from 395 – 1028 nm. The PAR between 400-700 nm should be later computed (mW/m2/sr/nm)
PARf – total incoming radiance spectrum from 395 – 1028 nm when the fluorescence filter is used. The PAR between 400-700 nm should be later computed (mW/m2/sr/nm)

Refl_real – real reflectance
Trans_real – real transmittance
Abs_real – real absorbance

Refl_real_smooth – real reflectance where between 656 and 658 nm the spectrum was smoothed to remove the small vias introduced when subtracting the fluorescence emission from the measured reflected radiance.
Trans_real_smooth – real transmittance where between 656 and 658 nm the spectrum was smoothed to remove the small vias introduced when subtracting the fluorescence emission from the measured reflected radiance.

Abs_real_smooth – real absorbance where between 656 and 658 nm the spectrum was smoothed to remove the small vias introduced when subtracting the fluorescence emission from the measured reflected radiance.

Fup – forward fluorescence (mW/m2/sr/nm)
Fdw – backward fluorescence (mW/m2/sr/nm)

Treatment – treatment time (c: control, s: water deficit, n: nitrogen deficit)
Repli – replicate number
Sample – sample number || L1x → pigments were analyzed from this sample.

¿How to call structure in MATLAB?

Write the structure name (PAR300, PAR1000) + "." + variable name

Example:
PAR300.wvl
PAR300.PAR
PAR300.Refl_real